

The Symbiotic Search Fabric

A mature end-to-end framework for co-adaptive, multi-coupled-tenant, client-owned, agent-operated, time-bounded search — in which the residue of agent experience is the raw material of the system's next design.

Classification. Internal — Strategy & Architecture. **Evidentiary status.** Notional throughout: this is a design synthesis, and no measured implementation exists. Where a mechanism has an Established analogue in the published field (anytime algorithms, differential privacy, secure aggregation, Shapley accounting, learning-coefficient drift estimation), the text marks it *adopt, not invent*. The evidentiary seam is preserved deliberately: nothing in this paper should be quoted as an achievement.

Abstract. This paper specifies a complete architecture for search systems that are operated by autonomous agents rather than people, owned by the clients whose corpora they index, coupled — never merged — across tenants, bounded in time at every interaction, and constitutionally rebuilt from the recorded experience of their own operators. The design is presented in strict order: first the ten obligations extracted from the original conceptualization; then a dedicated treatment of each plane and each of its components; only then, in a closed-world registry, the relationships among them; then the invariant lattice, a phased implementation program with explicit steps and validation burdens, and — as a dedicated part — a systematic account of how many other ways the same ten points could have been synthesized, and why this one was chosen.

Search here is not a service that agents call. It is a habitat they inhabit, own under charter, and rebuild from the record of their own foraging. Warp threads are sovereign tenant estates; weft threads are governed coupling channels; agents are the shuttles; and the loom re-strings itself, one adjudicated generation at a time.

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Part I — The problem, decomposed

1 · Reading note and evidentiary posture

This document is written to be verified, not admired. Three disciplines run through every section. First, components are introduced **before** any relationship among them is asserted: Part II names every part and sub-part of the architecture and states its function in isolation; Part III is the only place where edges between components are defined, and it does so under a closed-world convention — a relationship exists if and only if it appears in the Interface Registry (Table 3), and absence from the registry is prohibition, not omission. Second, every mechanism carries its evidentiary status: the whole paper is Notional, and mechanisms with Established analogues in the field are marked *adopt, not invent* so that novelty claims stay honest. Third, the paper treats its own design decisions as contingent: Part V enumerates the alternative syntheses of the same ten obligations, counts them, and states the selection rule under which this particular architecture — and not another — falls out.

2 · The original conceptualization: ten points

The originating sentence is quoted in full, because the decomposition must answer to it verbatim:

“Envision a mature end-to-end framework for co-adaptive, multi-coupled-tenant, client-owned, agent-operated, time-bounded search systems where agents’ experience shapes the evolving design.”

Read as an engineering brief, the sentence contains exactly ten load-bearing commitments. Each is stated below as an obligation together with the design question it opens; the architecture of Parts II-III is, clause by clause, the discharge of these obligations.

P1 · Maturity. The system is specified at governed steady state, not as a prototype that will acquire governance later. Rollback, audit, pricing, and exit are constitutive, present from the first day of operation. *Design question:* what must be true on day one for year-ten operation to be lawful?

P2 · End-to-endness. The framework closes its own loop: query admission, answer, deposited experience, and design change form one circulation with no seam that crosses an unspecified boundary. *Design question:* where, precisely, does the loop close, and what carries it across each joint?

P3 · Co-adaptivity. Agents adapt to the system and the system adapts to the agents. Adaptation is bidirectional by construction, not by accident. *Design question:* through what medium, at what feedback gain, and under whose control?

P4 · Tenant multiplicity. More than one principal inhabits the framework as a first-class citizen; tenancy is a unit of ownership and law, not a namespace prefix. *Design question:* what is the unit of tenancy, and what does it contain?

P5 · Tenant coupling. Tenants are coupled: value flows between them. The brief says *coupled*, not *pooled* — exchange with structure, not fusion. *Design question:* what may cross between tenants, at what price, and what may never cross?

P6 · Client ownership. The client owns the search estate — corpus, indexes, learned models, experience records, and the right to leave with all of it. *Design question:* what does ownership mean operationally: which keys, which export, which revocations?

P7 · Agent operation. Autonomous agents are the operators — they issue and refine queries, curate indexes, tune rankers, and audit outcomes. Humans hold authority but no operational verbs. *Design question:* what division of labor among agent roles, and what is irreducibly reserved to humans?

P8 · Time-boundedness. Every search carries a deadline contract, and every retained artifact carries a validity horizon. Time is a first-class term of service, not a performance aspiration. *Design question:* what happens — exactly — when the bound cannot be met?

P9 · Search as the organizing function. The workload is retrieval and foraging; the architecture is organized around what searching agents need rather than what human querying habits assume. *Design question:* what index geometry and memory stratification serve machine foraging?

P10 · Experience-shaped evolution. The design itself evolves, and the record of agent experience is the raw material — and the only legitimate warrant — for that evolution. *Design question:* how does lived trace become binding design without silent drift?

The ten points of the original conceptualization, and where each is discharged

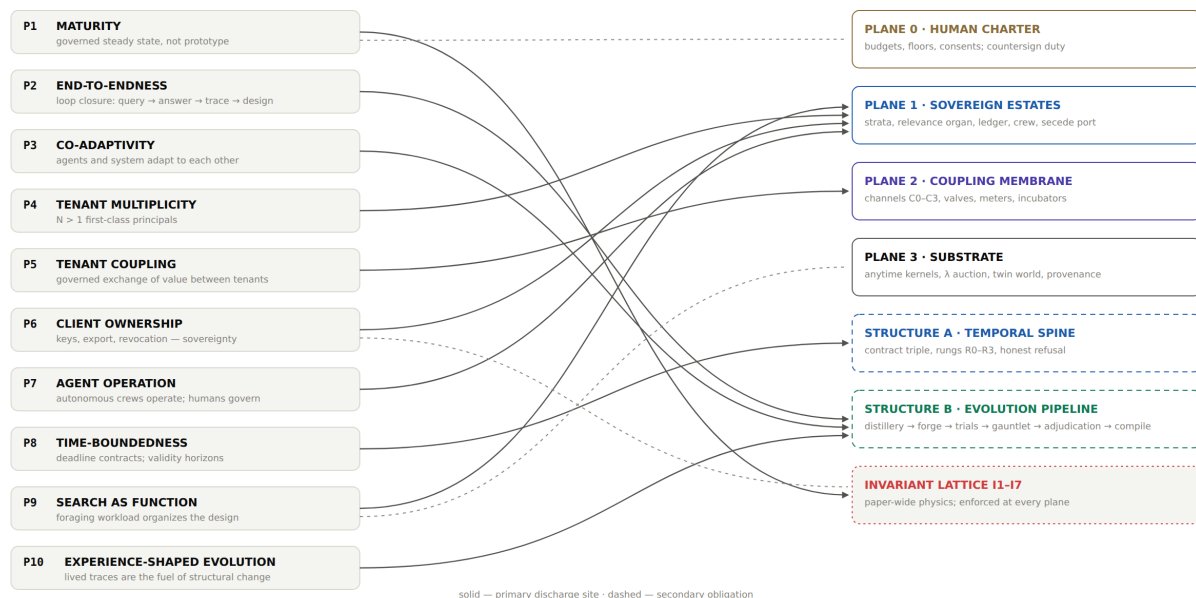


Figure 1. The ten points and their discharge sites. Solid lines mark the primary site where an obligation is discharged; dashed lines mark secondary obligations. Every point lands on a named plane, transverse structure, or the invariant lattice — none is discharged rhetorically.

3 · The governing rule and the shape of the answer

One rule governs the whole synthesis, and every later section can be derived from taking it seriously:

Share one admission calculus everywhere — value per unit cost against a posted price — but never confuse throughput with authority. Authority over the system's future is granted only by verification, and the burden of verification grows with a change's blast radius, not with its frequency.

The rule produces a deliberate asymmetry. Millions of queries per day are admitted by a cheap local test; a single design mutation per month is admitted by replayed trials, adversarial refutation, and countersigned adjudication. Between those extremes sit model updates (quarantined) and cross-tenant exchanges (metered). The architecture that carries this asymmetry consists of four horizontal planes — human charter, sovereign estates, coupling membrane, operator substrate — crossed by two transverse structures: a temporal contract spine through which all queries descend, and an evolution pipeline through which all design ascends. Three circulations tie them together: queries fall (blue), experience rises (amber), design returns (green).

A note on the frame drawn in Figure 2: the estates, the membrane, and their prices sit **inside** the generation frame $G(n)$ because they are products of design and therefore evolve; the human charter and the operator substrate sit **outside** it because they are parties to the design, providing respectively its legitimacy and its physics. This placement is not decorative — it states which parts of the world the evolution pipeline is permitted to rewrite.

Macro-architecture: four planes, two transverse structures, three circulations

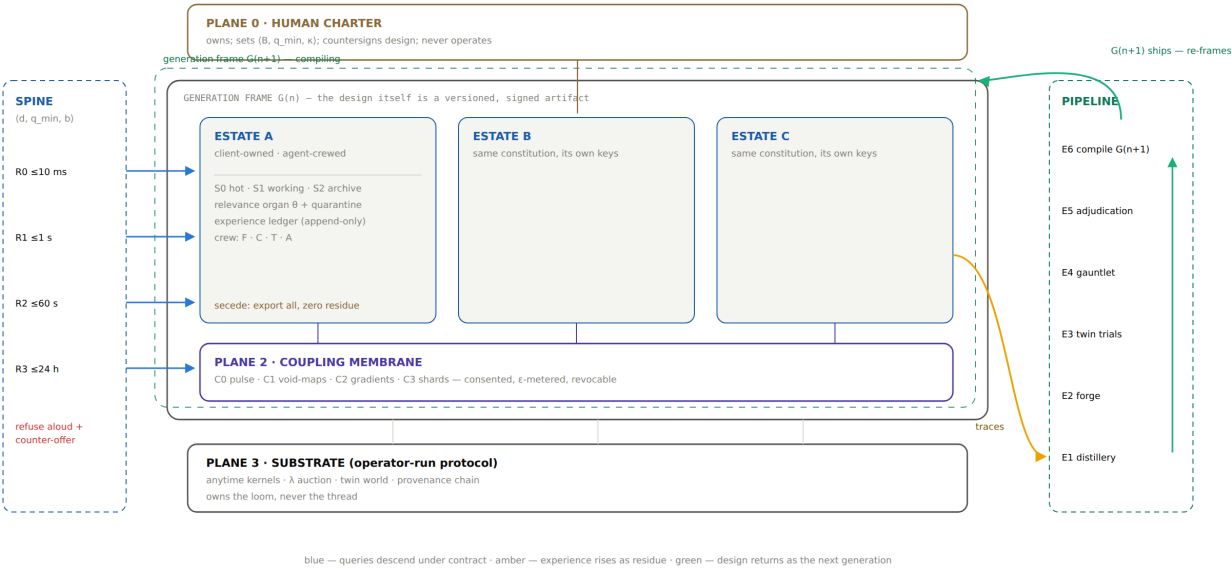


Figure 2. Macro-architecture. Four planes and two transverse structures, wrapped in a versioned generation frame: the design itself is a signed artifact, and the next frame is always compiling. The three circulations — query, experience, design — are the paper’s connective tissue and are specified edge-by-edge in Part III.

Part II — The planes, each in dedicated treatment

Convention: components are numbered within their plane (ES-2, MB-4, SB-3, SP-1, EV-6). Each section states the plane’s mandate, introduces every component and subcomponent with its function, and closes with the plane’s exclusions — what it may never do. No cross-plane relationship is asserted here beyond naming; all edges are defined once, in Part III.

4 · Plane 0 — The Human Charter Plane

Mandate. Plane 0 is where ownership and accountability live. Its occupants are people — the clients who own estates and the officers they appoint. The plane’s entire output is a small set of signed instruments; it performs no operations.

CH-1 · Charter Council. The standing body of client principals (one seat per tenant, plus appointed independent verifier seats). Function: to hold the signing keys of governance — approving charters, countersigning design generations, and seating disputes. Subcomponents: *seats* (identity-bound, non-delegable to agents), *quorum rules* (charter-defined, recorded), and the *countersign duty*: no design generation ships without council signature (see EV-5, §9).

CH-2 · Charter Instrument. The signed document each tenant issues; the only channel through which human intent enters the system. Function: to set the three operating quantities — the budget envelope *B* (what the tenant will spend), the quality floors *q_min* per query class (below which an answer is a breach, not a bargain), and the coupling consents *k* (which membrane channels are open, toward whom, at what privacy budget) — together with a liability posture for refusals and breaches.

CH-3 · Dissent Register. An append-only record of council votes and minority positions. Function: to make disagreement durable — dissent is logged, never overwritten, so that a later failure can be traced to the argument that predicted it.

Exclusions. Plane 0 has no operational verbs: it cannot issue queries, touch strata, read another tenant’s matter, or tune a model. Humans govern; they do not operate. This is the discharge of P7’s reserved-to-humans clause: legitimacy, not labor.

5 · Plane 1 — The Sovereign Tenant Estates

Mandate. Plane 1 is where all tenant matter lives and all searching happens. Each tenant occupies one estate: a cryptographically bounded territory containing corpus, indexes, models, experience, and the agent crew that works them. Estates are identical in constitution and disjoint in content — the same law, never the same matter. This plane discharges P4 (multiplicity), P6 (ownership), P7 (agent operation), and P9 (search-as-function).

ES-0 · Boundary and keys. The estate’s perimeter. Function: everything inside is tenant property under tenant keys *K_T*; every artifact that leaves is signed, every artifact that enters is quarantined (see MB-4). The boundary is what makes the word “owned” in the brief operational rather than merely contractual.

ES-1 · Corpus Vault. The client’s raw matter — documents, records, media — encrypted under client keys. Function: source of truth; nothing is searchable until a curation verb promotes it into the strata, so the vault can hold material the tenant never exposes to search at all.

ES-2 · Stratified Index (S0/S1/S2). The estate’s searchable memory, stratified by residence time rather than by data type — the architecture’s inheritance from metabolic-memory doctrine. **S0, the hot lattice** ($\tau \approx \text{ms}$): sketches, fingerprints, and caches able to answer in microseconds; its epistemic standard is sanity, not scholarship. **S1, the working index** ($\tau \approx \text{s}$): approximate-nearest-neighbor graphs and posting structures built for early-exit descent, where answer quality improves monotonically with time spent. **S2, the deep archive** ($\tau \approx \text{min-d}$): exhaustive, cold, reachable only by campaign-class queries. Function of the stratification: depth is purchased by deadline, never presumed (the spine, §8, enforces this); and each stratum carries its own validity predicate, so “correct” at ten milliseconds is a different claim than “correct” overnight. *Adopt, not invent*: anytime retrieval structures and early-exit ANN search are Established; their stratification under deadline law is the contribution.

ES-3 · Relevance Organ θ . The estate’s learned ranking model, client-signed. Subcomponents: **ES-3a, the drift gauge** — an online estimator of learning capacity (a λ -hat statistic in the spirit of learning-coefficient estimation; *adopt, not invent*) whose spikes signal that the model is undergoing a capability transition rather than a smooth improvement; and **ES-3b, the quarantine cell** — a holding stage through which every parameter delta must pass, releasing updates only between missions and only when the gauge is quiet (invariant I6). Function: relevance learning at full speed with structurally bounded surprise.

ES-4 · Experience Ledger. The estate’s append-only memory of its own operation, and the architecture’s single most consequential component: it is the sole legitimate fuel for design evolution (P10). Every search deposits a trace with six mandatory fields: the query contract as admitted, the descent path taken, the cost actually spent, the outcome delivered (including refusals), the regret estimate (what better answer later surfaced, if any), and the dead-ends and reformulation chain en route. Failures are first-class deposits: a dead-end atlas is an asset, not an embarrassment (invariant I4). A tail-event annex pins rare, expensive failures so they survive later compression.

ES-5 · Agent Crew. The estate’s operators — four charters, any number of instances. **The Forager** executes searches: it descends strata under contract, assembles anytime answers, and owns the deposit duty. **The Curator** works the strata: promotion and demotion between S0/S1/S2, cache seeding, vault-to-stratum admission, eviction hygiene. **The Tuner** works the organ and the wallet: it shapes θ within quarantine law and bids the estate’s budget at the spine’s gates. **The Auditor** attests: it verifies deposits, signs exports, and is the estate’s witness to the provenance chain. Clients may bring their own crew implementations; the charters, not the binaries, are the law.

ES-6 · Secede Port. The exit. Function: on unilateral demand, produce a complete signed export — vault, strata, θ , ledger, provenance extracts — revoke all couplings, and leave zero residue behind (invariant I1). The port is load-bearing precisely because it is never supposed

to be used: its verified existence is what makes every other promise credible.

ES-7 · The Fast Loop. Not a part but a licensed behavior: on timescales of seconds to hours, the crew retunes strategy — query plans, bids, cache policy, reformulation habits. Function: give co-adaptation its rapid, cheap half. The fast loop touches strategy only; by construction it cannot alter schema, strata geometry, API verbs, prices, or anything else that belongs to the design (that power lives only in Structure B, §9).

Plane 1 — anatomy of a sovereign tenant estate (components ES-0 ... ES-7)

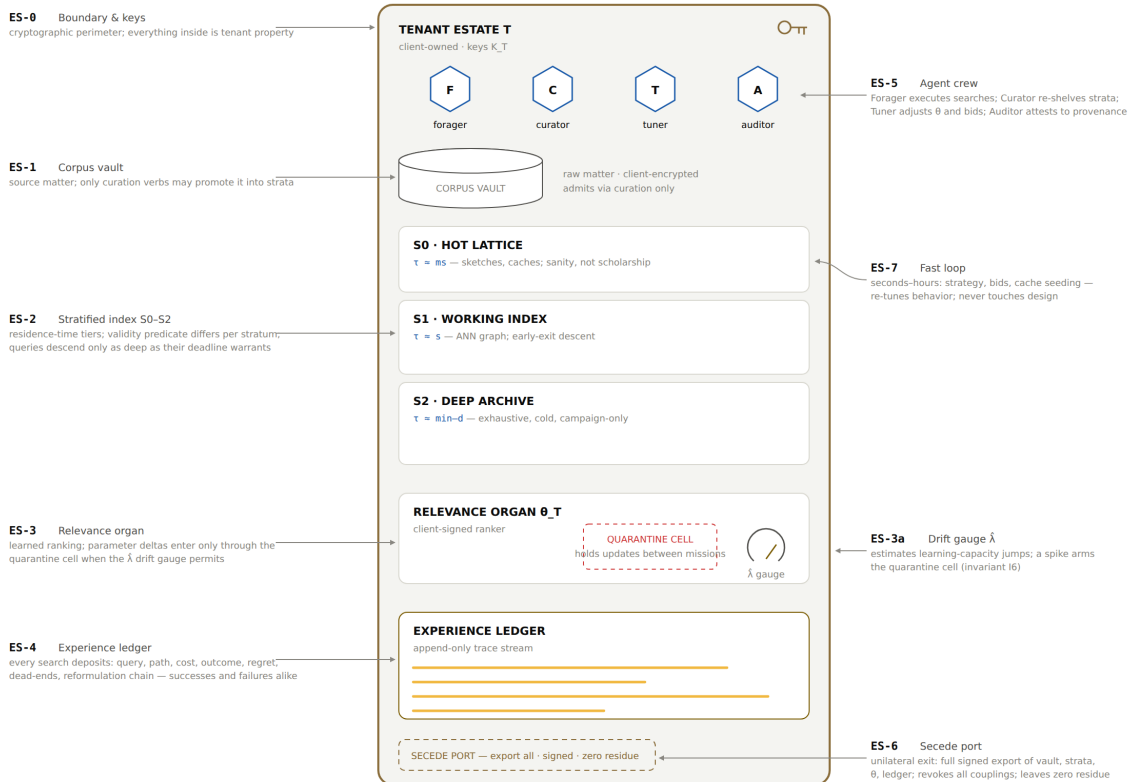


Figure 3. Anatomy of one estate. Components ES-0 through ES-7; callouts state each part’s function. All estates share this constitution; none shares its content.

Exclusions. An estate may not read another estate’s matter (all exchange is membrane-mediated), may not mutate its own design (fast loop is strategy-only), and may not silently discard experience (the deposit duty is unconditional).

6 · Plane 2 — The Coupling Membrane

Mandate. Plane 2 is the entire answer to the brief’s hardest phrase, “multi-coupled-tenant”: tenants must trade value without trading sovereignty. The membrane is a graded, consented, metered, revocable exchange surface between estates — coupled, never merged. It discharges P5 and, jointly with ES-0/ES-6, protects P6.

MB-1 · The four channels. Coupling is not one thing but a ladder of four, ordered by risk and by the consent strength required to open them. **C0 · Pulse** — differentially private telemetry aggregates (load, latency distributions, coarse hit rates). Always on; the federation’s weather report; lowest risk. *Adopt, not invent*: DP aggregation is Established. **C1 · Void-maps** — negative knowledge: dead-end atlases, exhausted-region markers, futile-reformulation warnings. Opt-in; remarkably high value per bit, because knowing where nothing lives rarely identifies what does. **C2 · Gradient federation** — signed model deltas exchanged under secure aggregation, letting relevance organs learn jointly without exposing traces (*adopt, not invent*). **C3 · Consortium shards** — leased, jointly searchable sub-indexes; the strongest coupling short of merger, under the strongest consent. Channel width in Figure 4 encodes this ladder.

MB-2 · Valves and consent keys κ . Every channel between every tenant pair carries a valve that opens only under a live, charter-issued consent key κ_{AB} . Function: make coupling an explicit, inspectable state rather than an emergent property of shared infrastructure.

MB-3 · ϵ -Meters. Privacy-budget accounting on every open valve: each exchange spends from a finite ϵ allowance set in the charter; an exhausted meter closes the valve until the council renews it. Function: leakage is bounded by arithmetic, not by hope.

MB-4 · Incubators. Quarantine chambers on the receiving side of every channel. Foreign signals — a void-map, a gradient delta, a shard update — sit in incubation while the estate’s auditor checks provenance and the drift gauge baseline. Function: the immune posture; nothing foreign touches an organ or a stratum directly.

MB-5 · Contribution Accounts. Shapley-style give-and-take accounting per tenant pair and per channel (*adopt, not invent*). Function: keep coupling reciprocal over time and make free-riding legible before it becomes resentment — the political failure mode of federations.

MB-6 · Revocation Lever. Unilateral, immediate, logged closure of any consent by its issuing tenant. On revocation the channel drains: in-flight exchanges complete or abort per charter, incubators flush, and the ϵ -meter freezes for audit. Function: consent that cannot be withdrawn instantly is not consent.

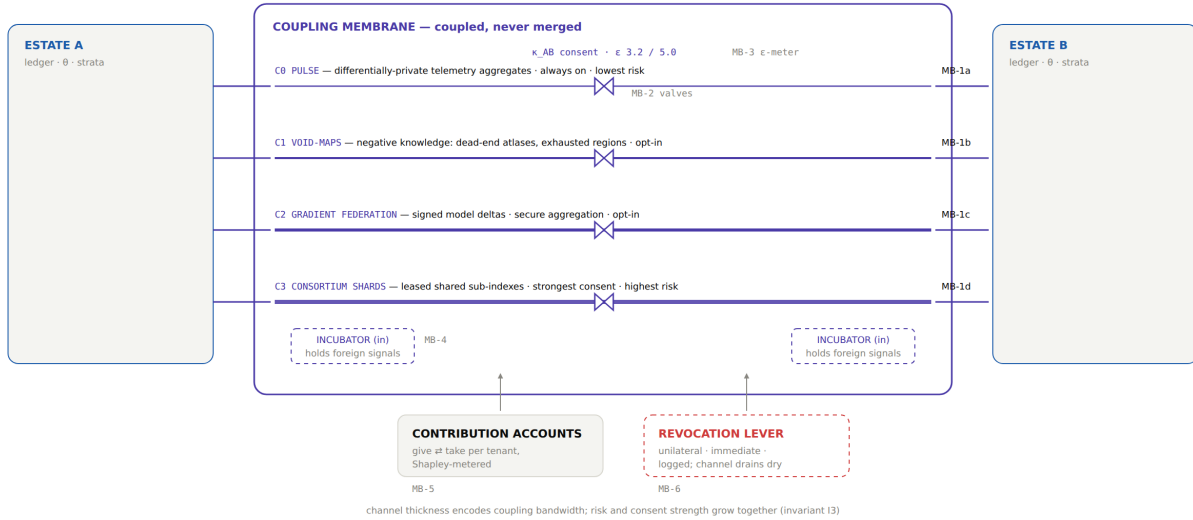
Plane 2 — the coupling membrane between any two estates (components MB-1 ... MB-6)

Figure 4. The membrane between any two estates. Four channels in rising order of coupling bandwidth and risk; valves under consent keys; ϵ -meters bounding leakage; incubators quarantining inbound signals; contribution accounts and the revocation lever.

Exclusions. The membrane carries derived signals only — aggregates, negative knowledge, deltas, leases. Raw corpus matter never crosses it on any channel; a tenant wanting another’s documents is asking for a contract, not a coupling.

7 · Plane 3 — The Operator Substrate

Mandate. Plane 3 is what the framework operator runs: the shared physics under all estates. Its one-sentence constitution: *the operator owns the loom, never the thread* — every substrate surface is protocol, and no substrate surface is tenant data.

SB-1 · Anytime Kernels. The library of interruptible algorithms every estate builds on: early-exit ANN descent, sketch-then-sweep ladders, progressive scoring. Function: guarantee that any computation can stop at deadline d and hand back the best answer so far, with quality monotone in time spent. This is the mechanical premise of the entire spine (§8). *Adopt, not invent:* anytime algorithms are Established; making them the only lawful kernel class is the design choice.

SB-2 · The λ Auction. The scheduler, expressed as one shadow price. Every claim on shared capacity — compute, recall, bandwidth — is admitted by the same local test: $\text{admit } \Delta \text{value} / \text{cost} \geq \lambda(t)$. The price rises under load and falls in slack; tuners bid estate budgets against it. Function: ration everything with one lever; make “the system is busy” a number rather than a mood. Estimation asymmetry is law: when uncertain, understate value and overstate cost, so ambiguity starves rather than floods the system.

SB-3 · The Twin World. A shadow replica of the federation — synthetic estates, recorded workloads, membrane simulacra — in which candidate designs are exercised. Function: the only arena where the evolution pipeline’s trials (EV-3) run; nothing tried in the twin has live

effect. The twin consumes traces under the same consents κ that govern the membrane.

SB-4 · The Provenance Chain. A hash-linked, third-party-auditable record of every consequential event: admissions, evictions, refusals, consents, revocations, quarantine releases, rollouts, secessions. Function: the paper trail that converts every promise in this document from assurance into evidence.

SB-5 · The Loom Protocol. The operator’s complete external surface: the wire formats, contract grammars, and proof obligations that estates and membrane implementations must satisfy. Function: substitutability — any module (including the operator’s own) is replaceable by any implementation honoring the protocol and its proofs, which is what keeps the operator disciplined.

Plane 3 — the operator substrate (components SB-1 ... SB-5)

SB-5 · THE LOOM PROTOCOL — the operator’s entire surface; tenant matter never crosses this line

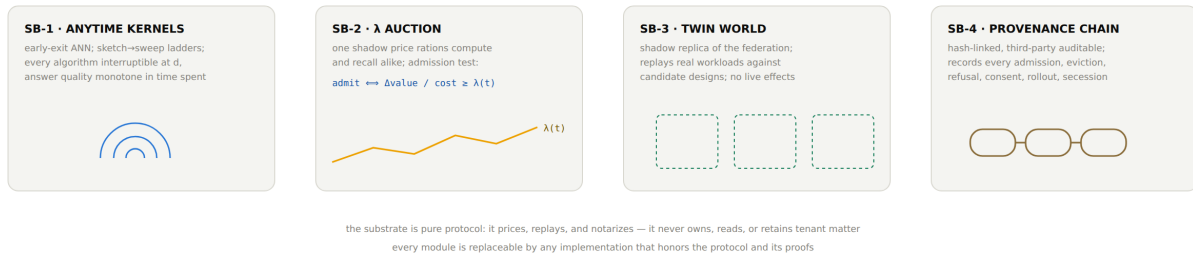


Figure 5. The substrate. Four modules beneath a single boundary rule: everything above the dashed line is tenant matter and never crosses it; everything below is protocol.

Exclusions. The substrate never owns, reads, retains, or prices tenant content as content; it prices claims on capacity. It cannot originate design proposals (it has no experience of its own) and holds no seat on the council.

8 · Structure A — The Temporal Contract Spine

Mandate. The spine is time-boundedness (P8) made structural: a single admission-and-descent path through which every query enters the fabric, carrying an explicit contract, and down which it may purchase depth. It is drawn vertical because that is what it is: a ladder of deadlines aligned to the strata of every estate.

SP-1 · Query Contract. Every query arrives as the triple $\langle d, q_{\min}, b \rangle$ — deadline, quality floor, budget. Function: convert the vague “fast and good” into three numbers the rest of the machine can enforce. Contracts are written by requesting agents (or standing charter defaults) and are binding on both sides: the fabric may not exceed d or undercut $q_{\min}$; the requester may not consume more than b .

SP-2 · Rungs R0-R3. Four deadline classes, each admitting specific strata and kernels. **R0 Reflex** (≤ 10 ms): S0 only — sketch probes and cache hits; the contract is sketch-or-silence, and no speculative depth is permitted. **R1 Interactive** (≤ 1 s): S0+S1 under early-exit

descent; the answer in hand improves monotonically until the deadline expires. **R2 Deliberative** (≤ 60 s): bounded S2 access, multi-hop foraging, reformulation chains within budget. **R3 Campaign** (≤ 24 h): full archive sweeps and — where consents permit — coupled recall across the membrane. Function: make “how deep may this query go” a consequence of its contract, not of its enthusiasm.

SP-3 · Admission Gates. At every rung, one local test: $\text{ADMIT} \iff \text{bid} \geq \lambda(t) \wedge E[q \mid d, b] \geq q_{\min}$ — the bid clears the substrate’s price and the expected quality (estimated from ledger history) clears the floor. Estimation errors round conservative. Function: keep both scarcity and honesty at the door.

SP-4 · Honest Refusal. When the gate’s quality test fails, the spine refuses aloud and returns a counter-offer (d', q') : the nearest contract it believes it can honor. The system may degrade to the floor or decline explicitly; it may never be silently late or silently shallow (invariant I2). Every refusal is a logged, first-class outcome — and a ledger trace, because refusals are experience too. Function: preserve trust under load; a fabric that lies about time will be routed around, then abandoned.

SP-5 · Result Stamps. Every answer leaves stamped $(\text{TTL}, \text{confidence}, \text{cost}, \text{provenance ref})$. Function: validity is explicit and expiring, never assumed — the consuming agent knows exactly how long the answer is warranted and what it cost to make.

Structure A — the temporal contract spine (components SP-1 ... SP-5)

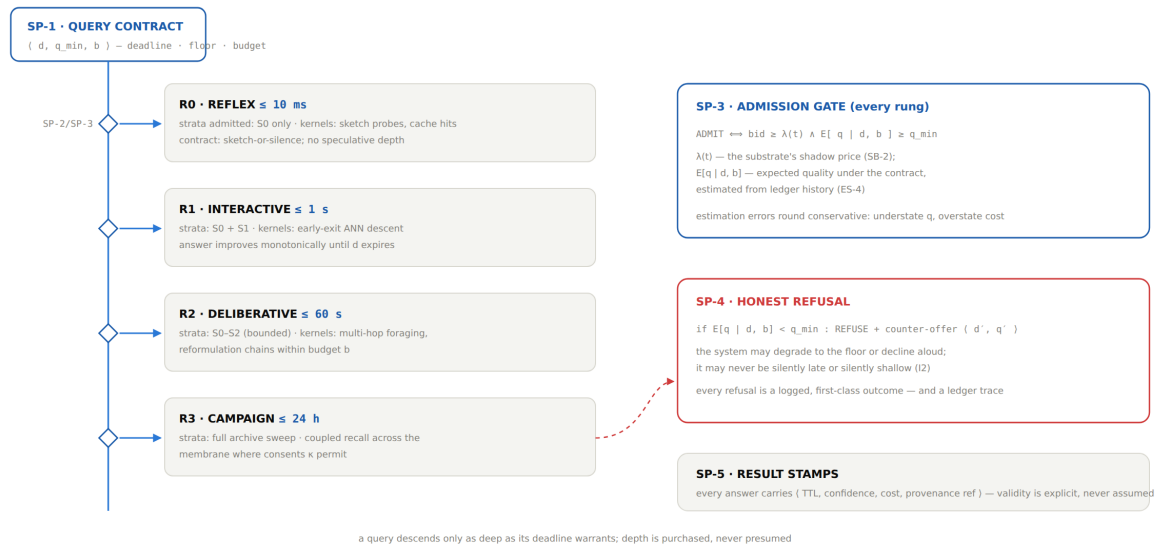


Figure 6. The temporal contract spine. A query’s contract admits it to a rung; each rung admits specific strata and kernels; the gate tests price and expected quality; failure produces an honest refusal with a counter-offer, never silence.

9 · Structure B — The Evolution Pipeline

Mandate. Structure B is P10 made governable: the six-stage path by which recorded experience — and nothing else — becomes binding design. It is the slow loop, deliberately weeks-per-cycle where the fast loop is seconds; the separation of timescales is load-bearing, because co-adaptation without timescale separation is resonance.

EV-1 · Distillery. Compresses ledger streams (scoped by consent) into decision-grade forms: regret spectra by query class, friction maps over strata and API verbs, refusal and dead-end statistics. Maintains the tail-event vault so rare failures survive compression. Function: turn millions of traces into a shape a proposal can cite.

EV-2 · Proposal Forge. Drafts design mutations: strata geometry, schema fields, API verbs, valve types, price schedules, rung boundaries. The forge's law is invariant I5's first clause: *no proposal without cited traces* — every mutation names the distillate evidence that motivates it. Function: keep design change answerable to lived experience rather than to fashion.

EV-3 · Twin Trials. Candidate designs replay recorded workloads in the twin world (SB-3), including counterfactual load and adversarial traffic mixes. Function: measure a proposal's consequences before any tenant feels them.

EV-4 · Gauntlet. Independent adversarial verifiers attempt refutation: leakage across κ boundaries, regression against quality floors, capture of the proposal by the metric it cites. Refuters default to "refuted" when uncertain. Function: institutionalized disbelief between measurement and law.

EV-5 · Adjudication. The charter council (Plane 0) countersigns surviving proposals under its quorum rules; dissent enters the register. Function: the constitutional joint — the one point where human authority touches design, and the discharge of P1's legitimacy clause.

EV-6 · Generation Compiler. Compiles the countersigned change-set into the next generation frame $G(n+1)$: staged rollout, rollback rail armed, signed release. Function: design change as an atomic, reversible, attributable event — never a drift.

EV-7 · Resonance Damper. The governor on the whole double loop: release requires the estimated co-adaptive loop gain — how strongly design responds to experience times how strongly experience responds to design — to stay below one (invariant I7). Function: co-adaptation is the point; runaway co-adaptation is the failure mode. The damper is what lets the fabric court the first without suffering the second.

EV-8 · Fossil Record. Refuted proposals and retired organs, kept legible with their evidence and dissents. Function: institutional memory — the API is a record of struggle, and a rejected idea returns only against the argument that killed it.

EV-9 · Generation Frames. The versioned design artifacts themselves: $G(n)$ live and signed, $G(n+1)$ compiling, predecessors retired to fossils. Function: make "the design" a noun with a hash, so that P10's "evolving design" is a sequence of auditable objects rather than a mood.

Structure B — the experience→design evolution pipeline (components EV-1 ... EV-9)

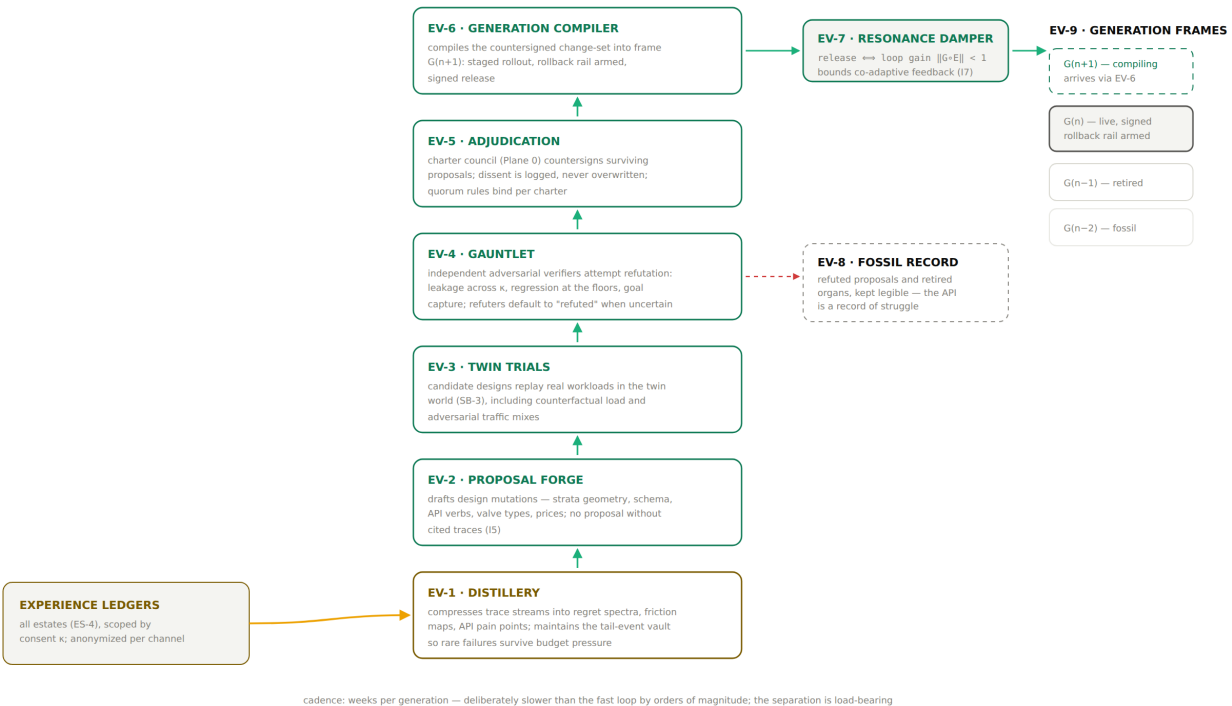


Figure 7. The evolution pipeline. Experience ascends through distillation, proposal, trial, refutation, and adjudication before compiling into a signed generation; the damper bounds the loop gain; fossils keep every refusal legible.

Part III — Relations, stated once and closed

10 · The three circulations and the Interface Registry

Components having been introduced, their relationships can now be stated under a closed-world convention: **an interface exists if and only if it appears in Table 3**. Absence is prohibition. Three composite circulations give the registry its narrative shape. The **query circulation** descends: contract → gate → strata → stamped answer (IF-04, -05, -06, -26). The **experience circulation** ascends: deposit → ledger → distillery (IF-07, -08, -18). The **design circulation** returns: distillate → proposal → trial → gauntlet → adjudication → compiled generation → every plane (IF-19 through IF-24). Everything else is governance and sovereignty plumbing: charter instruments flowing down, consents and revocations flowing sideways, provenance flowing always.

Principal interfaces — nodes are components, numbered edges key to the Interface Registry (Table 3)

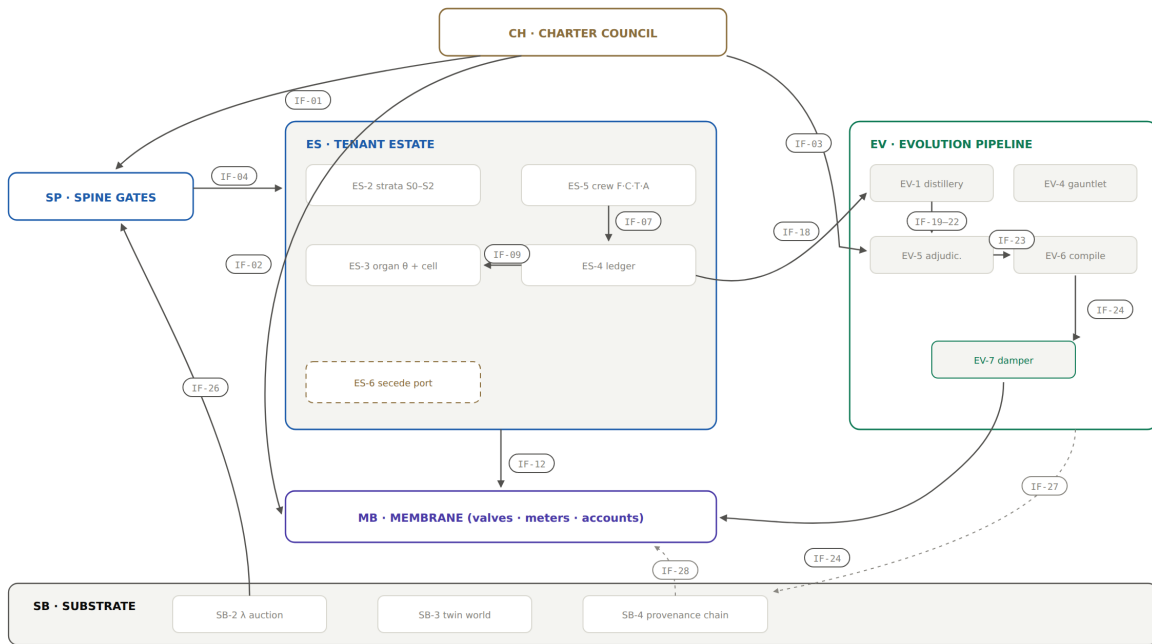


Figure 8. Principal interfaces. Nodes are components from Part II; numbered edges key into Table 3. The drawing shows the principal edges only; the table is normative and complete.

Table 3 · The Interface Registry (normative; closed-world).

ID	Source → Sink	Payload	Cadence	Gate / Price	Inv.
IF-01	CH-2 → SP-3	floors $q_{\min}$, budgets B	on charter change	countersigned	I2
IF-02	CH-2 → MB-2	consents κ (open/scope)	on charter change	countersigned	I3
IF-03	CH-1 → EV-5	countersignature / dissent	per proposal	quorum	I5
IF-04	SP-3 → ES-2	admitted query (contract)	per query	$\text{bid} \geq \lambda \wedge E[q] \geq q_{\min}$	I2
IF-05	SP-4 → requester	refusal + counter-offer $\langle d', q' \rangle$	on gate failure	logged	I2

ID	Source → Sink	Payload	Cadence	Gate / Price	Inv.
IF-06	ES-2/ES-5 → requester	stamped answer (TTL,conf,cost,prov)	per query	stamp mandatory	I2
IF-07	ES-5 → ES-4	trace deposit (6 fields)	per search, incl. refusals	unconditional	I4
IF-08	ES-4 → ES-5 (Tuner)	regret batches	fast loop (s-h)	read-only	I4
IF-09	ES-5 (Tuner) → ES-3	θ parameter delta	between missions	via ES-3b; gauge quiet	I6
IF-10	ES-5 (Curator) → ES-2	promotion / demotion / eviction	continuous	curation verbs only	I4
IF-11	ES-5 (Auditor) → SB-4	attestations	continuous	signed	I5
IF-12	ES-4 → MB-1a (C0)	DP telemetry aggregates	always on	ϵ -metered	I3
IF-13	ES-x $\rightleftharpoons$ MB-1b (C1)	void-maps (negative knowledge)	per exchange	κ + incubation	I3, I6
IF-14	ES-3 $\rightleftharpoons$ MB-1c (C2)	signed model deltas	per epoch	secure agg + incubation	I3, I6
IF-15	ES-2 $\rightleftharpoons$ MB-1d (C3)	shard leases	per lease	strongest κ	I1, I3
IF-16	MB-1 → MB-5	give/take entries	per exchange	Shapley metering	I3
IF-17	tenant → MB-6	revocation of κ	anytime	unilateral, immediate	I1, I3
IF-18	ES-4 (all) → EV-1	trace batches	slow loop (d-wk)	scoped by κ ; anonymized	I4, I5
IF-19	EV-1 → EV-2	distillates (regret spectra ...)	per cycle	citable forms only	I5
IF-20	EV-2 → EV-3	proposal bundle + cited traces	per proposal	twin-only execution	I5
IF-21	EV-3 → EV-4	trial results	per proposal	adversarial handoff	I5
IF-22	EV-4 → EV-5	surviving proposals + refutation log	per proposal	default-refuted	I5
IF-23	EV-5 → EV-6	countersigned change-set	per generation	quorum met	I5
IF-24	EV-6 → all planes	generation G(n+1)	staged rollout	damper gain < 1; rollback armed	I5, I7
IF-25	EV-4 → EV-8	refuted proposals + evidence	per refutation	archived legible	I5
IF-26	SB-2 → SP-3	shadow price $\lambda(t)$	continuous	none — it is the price	I2
IF-27	EV-3 $\rightleftharpoons$ SB-3	replayed workloads	per trial	sandboxed; consented traces	I5
IF-28	all components → SB-4	provenance appends	always	hash-linked	I1, I5
IF-29	ES-6 → departing tenant	full signed export; couplings revoked	on demand	zero residue verified	I1

Reading the registry, three properties deliver the “minimal ambiguity” the design owes its reader. **Completeness:** every arrow in every figure corresponds to a row here; a relationship not listed does not exist and must not be implemented. **Directionality:** payloads move only as written; the membrane rows (IF-12 through IF-15) are the only bidirectional edges in the fabric, and each is gated on both sides. **Guarded authority:** the two edges that can change what the system *is* — IF-09 into a live model and IF-24 into a live design — are exactly the two carrying quarantine and damper conditions. Throughput edges are many and cheap; authority edges are few and expensive. That asymmetry is §3’s governing rule, now visible as

table structure.

11 · The invariant lattice I1-I7

Seven invariants bind every plane; each names its enforcing components. **I1 · Sovereignty.** Every byte of tenant state is exportable and client-signed; secession leaves zero residue. *Enforced by* ES-0, ES-6, MB-6, SB-4 (IF-17, -28, -29). **I2 · Honest time.** Meet the deadline, degrade to the floor, or refuse aloud with a counter-offer; never silently late, never silently shallow. *Enforced by* SP-1...SP-5, SB-1, SB-2 (IF-04, -05, -06, -26). **I3 · Graded coupling.** Coupling bandwidth is earned, consented, ϵ -metered, and unilaterally revocable; it widens only with demonstrated benefit and bounded leakage. *Enforced by* MB-1...MB-6 (IF-12...-17). **I4 · Experience is capital.** Every search deposits a complete trace; dead-ends and refusals are assets; the deposit duty is unconditional. *Enforced by* ES-4, ES-5 (IF-07, -08, -10, -18). **I5 · Design provenance.** No design mutation without cited traces, a twin trial, an adversarial gauntlet, and countersigned adjudication. *Enforced by* EV-1...EV-6, EV-8, CH-1, SB-4 (IF-19...-25, -28). **I6 · Drift quarantine.** Model-capability jumps are held between missions until verified; foreign signals incubate before absorption. *Enforced by* ES-3a, ES-3b, MB-4 (IF-09, -13, -14). **I7 · Bounded resonance.** The co-adaptive loop wears a damper: estimated agent $\Rightarrow$ design feedback gain stays below one at every release. *Enforced by* EV-7 (IF-24).

The lattice has an internal order that Part V will exploit: I1 and I2 are lexical — no quantity of benefit purchases their violation; I3-I6 are protective — they bound the mechanisms that make the fabric profitable; I7 is reflexive — it protects the system from its own central virtue.

Part IV — Implementation: phases, steps, validation burdens

12 · The build program

The program is ordered by the governing rule: substrate physics first, one sovereign estate second, federation third, learning-federation fourth, generational maturity last. Each phase lists its build steps, then the validation burden that must be discharged — by demonstration, not assertion — before the next phase may begin.

Phase 0 · Substrate and calculus (foundation)

Steps. (1) Implement SB-4, the provenance chain, first — nothing else may come online unwitnessed. (2) Implement SB-2, the λ auction, with the conservative rounding law built into its estimators. (3) Assemble SB-1, the anytime kernel library, rejecting any algorithm that cannot prove interruptibility and monotone answer quality. (4) Implement the spine skeleton SP-1...SP-5 against a stub estate: contracts, gates, refusal with counter-offer, stamps. (5) Define the trace grammar (the six mandatory fields of ES-4) as a wire standard. (6) Stand up SB-3, the twin world, as a replay harness over synthetic workloads. (7) Freeze SB-5, the loom protocol, as the versioned boundary all later work programs against. **Validation burden.** Adversarial latency injection cannot make a gate silently late (I2 holds under fault); the auction starves ambiguous bids (safe-side rounding demonstrated); provenance chain survives third-party audit replay. **Exit criterion.** A stub query descends, is refused honestly at an unsatisfiable contract, and both outcomes appear on the chain.

Phase 1 · One sovereign estate (single-tenant operation)

Steps. (1) Erect ES-0 boundary and key ceremony; client holds K_T from the first hour. (2) Load ES-1 vault; implement curation verbs. (3) Build strata S0/S1/S2 and bind them to rungs R0-R2. (4) License the crew charters ES-5; run foragers and curators live, tuners in shadow. (5) Enforce the deposit duty — every search, including refusals, writes a complete trace. (6) Bring ES-3 online with the drift gauge recording and the quarantine cell enforcing between-mission release. (7) Drill ES-6: full secession export to cold storage, verified restorable. **Validation burden.** Deposit completeness $\approx 100\%$ under load; quarantine provably interposes on every θ delta (attempted hot update fails closed); the secede drill completes within its contractual bound and leaves zero residue by independent inspection. **Exit criterion.** Thirty days of single-estate operation with I1, I2, I4, I6 continuously green on the chain.

Phase 2 · Federation (coupling without learning)

Steps. (1) Onboard estates two and three under identical constitutions. (2) Open C0 pulse with ε -meters live from the first byte. (3) Stand up MB-5 contribution accounts in shadow. (4) Open C1 void-maps behind incubators; measure forage-efficiency lift attributable to shared negative knowledge. (5) Rehearse MB-6 revocation quarterly: close a live consent, verify drain semantics and meter freeze. (6) Extend R3 campaign contracts to consented cross-tenant recall. **Validation burden.** ε accounting exact under audit replay; revocation drains within bound; no cross-estate influence path exists outside membrane rows of the

registry (verified by information-flow test). **Exit criterion.** Demonstrated C1 benefit with bounded, metered leakage — the empirical warrant I3 requires before any wider channel opens.

Phase 3 · Learning federation (the pipeline, gated)

Steps. (1) Open C2 gradient federation under secure aggregation, incubators mandatory. (2) Bring EV-1 distillery online over consented ledgers; publish regret spectra to tenants. (3) Activate EV-2...EV-4 against the twin world only. (4) Seat the council formally (CH-1...CH-3); run EV-5 on the first surviving proposals. (5) Ship the first micro-generation through EV-6 — a deliberately small change (one API verb, one price revision) to exercise rollout and rollback rails end-to-end. **Validation burden.** Every shipped mutation cites traces (I5 audit); gauntlet demonstrates teeth by refuting a majority of first-round proposals; rollback executes within bound on a staged failure. **Exit criterion.** One full experience→design cycle on the chain, dissent register non-empty (a council that has never dissented has not yet governed).

Phase 4 · Generational maturity (co-adaptation under damping)

Steps. (1) Move to a regular generation cadence (weeks). (2) Bring EV-7 damper estimation live; publish loop-gain estimates with every release. (3) Open C3 consortium shards for tenants whose contribution accounts support them. (4) Operationalize the fossil record's return rule — a rejected proposal re-enters only against its killing argument. (5) Run annual secede-and-restore drills per tenant; publish results. **Validation burden.** Damper gain measurably below one across three consecutive generations; refusal rates at fixed floors non-increasing generation over generation (the design loop is reducing, not gaming, regret); tail-event retention demonstrated under budget pressure. **Exit criterion.** None — this is the operating state. Maturity (P1) is not a milestone but the condition in which every prior burden stays continuously discharged.

Part V — The other ways: alternative syntheses of the ten points

13 · Counting the design space, then quotienting it

A specification that cannot say what else it could have been is advocacy, not architecture. This part treats the ten points of §2 as free variables, enumerates the architectural stances each admits, counts the raw combinations, prunes the incompatible, and collapses the remainder into archetype families — then states the selection rule under which the Fabric, and not one of its rivals, is the answer. All counts are illustrative arithmetic over a stated enumeration, not measurements; their role is to make “how many other ways?” a question with a defensible shape rather than a shrug.

13.1 · The stance matrix

Point	Admissible stances (chosen ★)	k
P1 maturity	prototype-first · retrofit-governance · constitutional ★	3
P2 closure	open pipeline · dual-loop ★ · triple-loop (meta-governor)	3
P3 co-adaptivity	parametric-only · protocol-negotiated · structural, gated ★ · unconstrained-emergent	4
P4 tenancy unit	namespace · account-in-market · sovereign estate ★ · organ-of-organism	4
P5 coupling topology	broker-mediated · pairwise membrane ★ · hub consortium · full-mesh market	4
P6 ownership	custodial · escrowed · sovereign-with-export ★ · self-hosted-protocol	4
P7 operation	single monolithic agent · homogeneous swarm · differentiated crew ★ · open agent market	4
P8 time regime	soft SLA · deadline ladder ★ · continuous-price-only · per-query negotiation	4
P9 search substrate	index-centric strata ★ · model-internalized (RAG-absorbed) · graph-centric · hybrid	4
P10 evolution	vendor-offline redesign · continual online learning · generational, gated ★ · population-evolutionary	4

Table 4. Each point admits three or four defensible architectural stances. The enumeration is conservative — stances that differ only in vocabulary are merged.

13.2 · Raw count, compatibility pruning, equivalence quotient

The raw space is the product of the stance counts: $3 \cdot 3 \cdot 4^8 = 589,824$ distinct stance-vectors. Not all are coherent. Six compatibility rules do most of the pruning. **R-a:** unconstrained-emergent co-adaptation (P3) is inadmissible under a constitutional maturity posture (P1) — one cannot guarantee what one refuses to bound. **R-b:** organ-of-organism tenancy (P4) collapses the coupling axis (P5): organs do not trade, they share; three of four P5 stances vanish. **R-c:** custodial ownership (P6) removes the client’s standing to countersign, striking gated-generational evolution (P10) from its row. **R-d:** continuous-price-only time (P8) cannot express quality floors and is incompatible with dual-loop closure (P2), which needs floors to define regret. **R-e:** model-internalized search (P9) under sovereign-with-export ownership (P6) forces whole-model export rights, striking escrowed hybrids. **R-f:** open agent markets (P7) under constitutional maturity (P1) require a licensing layer that collapses them, in effect, into differentiated crews. Applied jointly, the

rules eliminate roughly four of every five vectors, leaving on the order of 1.2×10^5 admissible configurations — still absurdly many, which is the point: stance-vectors are not designs.

The meaningful count comes from a quotient. Call two configurations equivalent when they differ only by plane-merging or plane-splitting, by renaming, or by dual representation of the same governance relation (a valve “in the membrane” versus a filter “at the estate wall” is one mechanism drawn twice). Under this equivalence the admissible space collapses into approximately ten families, of which seven are load-bearing — internally coherent, defensible end-to-end, and genuinely distinct in what they optimize and what they sacrifice. Those seven are the subject of the next section. The honest summary of “how many different other ways”: **589,824 raw syntheses; about 1.2×10^5 jointly admissible; seven that matter.**

13.3 · The seven archetype families

A · The Fabric (chosen; this paper). Federal physiology: sovereign estates, graded membrane, deadline spine, gated generational evolution. Buys the forbidden corner — high coupling at high sovereignty — at the price of heavy governance machinery. Disqualifier it avoids: none within its own priorities; its risk is apparatus weight, examined in §14.

B · The Exchange. Market-first synthesis: every interaction — retrieval, coupling, even design change — is an auction; coupling is trade; tenancy is an account. Buys allocative efficiency and effortless prioritization under load. Forfeits floors: when everything is priced, quality floors and honest time become price-contingent, and a rich adversary can starve a poor tenant’s deadlines lawfully. Disqualified because I2 must be lexical, not purchasable.

C · The Commons. Ecological synthesis: one shared mycelial index; ownership as cryptographic tagging and policy filters rather than territory. Buys maximal learning velocity — every trace fertilizes every tenant. Forfeits hard sovereignty: export is possible but excision is not; a departing tenant’s influence remains diffused through shared structure forever. Disqualified on P6: exit without residue is unachievable by construction.

D · The Guild Blackboard. Classical multi-agent synthesis: one shared blackboard; role-guilds negotiate posts and claims; design evolves by consensus editing. Buys transparency and architectural simplicity. Forfeits both isolation and time-boundedness at scale — negotiation latency is unbounded, and the blackboard is one tenancy in all but name. Disqualified on P4 and P8 jointly.

E · The Archipelago. Isolation-first synthesis: estates as islands; coupling only through episodic, human-approved embassy missions carrying vetted parcels. Buys near-absolute sovereignty and auditability. Forfeits co-adaptation: experience pools too slowly for the design loop to learn anything before the world changes. Disqualified on P3/P10 starvation — the brief demands a living loop, not an annual summit.

F · The Organism. Total integration: tenants as organs of one body — single ledger, single relevance organ, unified crew. Buys the strongest co-adaptation physically available; nothing couples like shared flesh. Forfeits ownership entirely: secession is surgery, with scars on both sides. Disqualified on P6’s exit test — the first tenant to leave would falsify the architecture.

G · The Protocol Mesh. Operator-less synthesis: the substrate dissolved into open standards; estates self-host; evolution by versioned RFC. Buys credible neutrality — no operator to capture. Forfeits enforcement: invariants become conventions, the twin world has no owner, and the verification-burden asymmetry of §3 has no enforcer. Disqualified on P1: a mature system must be able to make its own promises binding.

The design space of the ten points — seven archetype syntheses (bubble area ≈ co-adaptation gain)

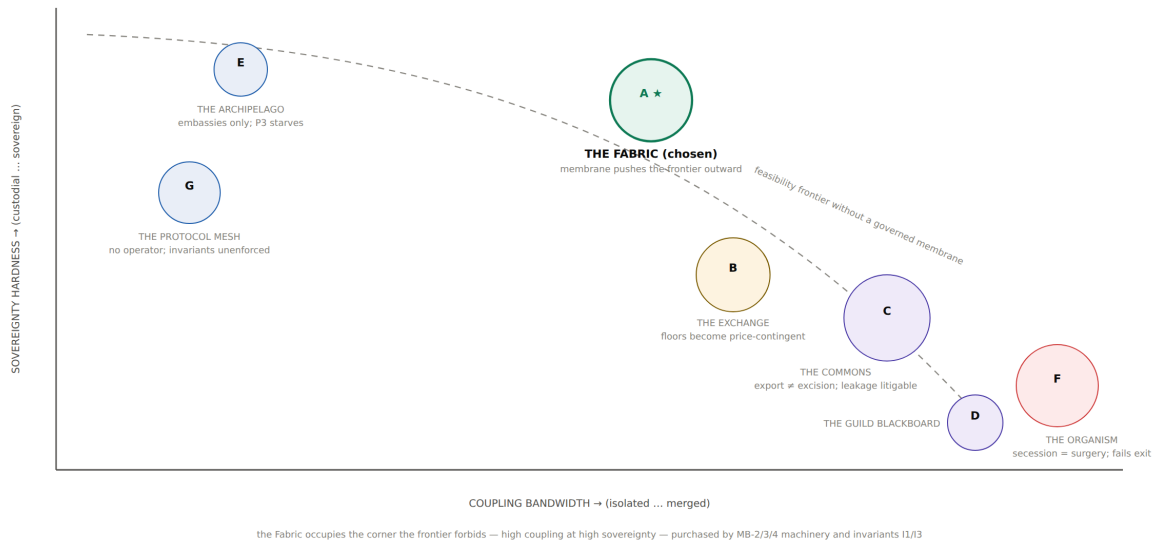


Figure 9. The seven archetypes on the two axes the brief cares most about. The dashed frontier is the trade-off available without governed-coupling machinery; the Fabric's wager is that membrane components MB-2/3/4 and invariants I1/I3 push that frontier outward.

13.4 · The selection functional

What actually chooses among the seven is not taste but priority order. Give the ten points a lexical ordering and maximize co-adaptation subject to everything above it, and each archetype appears as the optimum of a different ordering: put price efficiency first and B follows; learning velocity first and C follows; coherence first and F; neutrality first and G; auditability first and E. The Fabric is what falls out of exactly one commitment: $\{P6, P8\} > \{P3, P10\} > \text{efficiency}$ — sovereignty and honest time are lexical, co-adaptation is then maximized subject to them, and efficiency spends whatever freedom remains. There are $10! = 3,628,800$ strict priority orderings, but they collapse onto this same handful of basins — which is the deep reason “how many other ways?” has the answer it does: **combinatorially many syntheses, and only about seven kinds.** Choosing an architecture is choosing which points may never be traded away; everything else in this paper is consequence.

Part VI — Falsifiability and positioning

14 · What would prove this design wrong

Four falsifiers, stated against the paper's own claims. **First**, if metered coupling never demonstrates benefit — if C1/C2 lift over a C0-only baseline is statistically indistinguishable from zero in twin-world replay — then the membrane is apparatus without payoff and the fabric degenerates, honorably, into the Archipelago. **Second**, if the resonance damper must hold loop gain far below one to keep the federation stable, then generational co-adaptation is theater: the system is safe only because it barely co-adapts, and archetype E was the honest choice. **Third**, if refusal rates at fixed floors rise generation over generation, the design loop is optimizing something other than the regret it cites — Goodhart at the forge — and I5's citation discipline has failed its purpose. **Fourth**, if a secede drill cannot complete within its contractual bound, sovereignty is decorative and the architecture's central boast is false. Each falsifier is measurable inside the Phase 2-4 validation program; none requires waiting for a disaster.

15 · Positioning statement

Build a search fabric whose tenants are sovereign estates operated entirely by agent crews; whose every query carries a deadline contract enforced by honest refusal; whose tenants are coupled through a graded, consented, metered, revocable membrane and never merged; whose operator owns the loom and never the thread; and whose design is a signed, versioned artifact that only recorded experience — distilled, tried, refuted, and countersigned — may rewrite, under a damper that keeps co-adaptation below resonance. Verification burden grows with blast radius. Sovereignty and honest time are never traded. Everything else is negotiable, priced, and logged.

Appendix A — Glossary

Term	Definition
anytime kernel	an algorithm that can be interrupted at any moment and return its best answer so far, with quality monotone in time spent (SB-1)
archetype family	an equivalence class of stance-vectors that survive compatibility pruning and differ only in inessential structure (§13)
charter instrument	the signed tenant document fixing budgets B , floors $q_{\min}$, and consents κ (CH-2)
consent key κ	the live, charter-issued credential that opens a membrane valve between two tenants (MB-2)
contribution account	Shapley-metered give/take balance per tenant pair and channel (MB-5)
counter-offer $\langle d', q' \rangle$	the nearest honorable contract returned with every refusal (SP-4)
deposit duty	the unconditional obligation that every search, including refusals, writes a complete trace (ES-4, I4)
distillate	a decision-grade compression of ledger traces: regret spectra, friction maps, refusal statistics (EV-1)
drift gauge	the online estimator (λ -hat) of model learning-capacity whose spikes arm quarantine (ES-3a)
estate	one tenant's sovereign territory: vault, strata, organ, ledger, crew, ports (Plane 1)
fast loop	seconds-to-hours strategy adaptation by the crew; never touches design (ES-7)
fossil record	the legible archive of refuted proposals and retired organs (EV-8)
generation frame $G(n)$	the versioned, signed design artifact in force; the object the pipeline rewrites (EV-9)
honest refusal	declining a contract aloud, with counter-offer, rather than being silently late or shallow (SP-4, I2)
incubator	the receiving-side quarantine chamber on every membrane channel (MB-4)
loom protocol	the operator's complete external surface: formats, grammars, proof obligations (SB-5)
loop gain	the product of design-to-experience and experience-to-design sensitivities; released designs require it below one (EV-7, I7)
quarantine cell	the holding stage through which every model delta passes between missions (ES-3b, I6)
residence time τ	how long an artifact remains action-relevant; the stratification variable of ES-2
secede port	the verified unilateral exit: full signed export, revoked couplings, zero residue (ES-6, I1)
shadow price $\lambda(t)$	the single posted price rationing all claims on shared capacity (SB-2)
stance	one defensible architectural position on one of the ten points (§13.1)
trace	the six-field record a search deposits: contract, path, cost, outcome, regret, dead-ends/reformulations (ES-4)
twin world	the shadow replica in which candidate designs replay recorded workloads (SB-3)
void-map	shared negative knowledge: where searching has been shown futile (MB-1b)

Appendix B — Symbols

Symbol	Meaning
$\langle d, q_{\min}, b \rangle$	query contract: deadline, quality floor, budget (SP-1)
$\langle d', q' \rangle$	counter-offered contract accompanying a refusal (SP-4)
$\lambda(t)$	shadow price of shared capacity at time t (SB-2)
λ -hat	online drift/learning-capacity estimate on the relevance organ (ES-3a)

Symbol	Meaning
ϵ	privacy budget spent per membrane exchange; finite per consent (MB-3)
κ, κ_{AB}	coupling consent; the $A \leftrightarrow B$ instance (MB-2)
θ, θ_T	relevance-organ parameters; tenant T's instance (ES-3)
τ	residence time of an artifact in a stratum (ES-2)
$S_0 / S_1 / S_2$	hot lattice / working index / deep archive (ES-2)
$R_0\text{--}R_3$	deadline rungs: reflex, interactive, deliberative, campaign (SP-2)
$G(n)$	the n-th generation frame; $G(n+1)$ is always compiling (EV-9)
$\ G \circ E\ $	co-adaptive loop gain; release requires < 1 (EV-7, I7)
$E[q \mid d, b]$	expected answer quality under a contract, estimated from ledger history (SP-3)
B	tenant budget envelope fixed in the charter (CH-2)

Prepared as a design synthesis against the standards of the relevant disciplines (multi-agent systems, information retrieval, federated learning, mechanism design, functional-safety governance). Every mechanism marked adopt-not-invent has Established literature behind it; every count in Part V is illustrative arithmetic over a stated enumeration; and where this paper asserts more than a notional design can warrant, the reader is invited — per §14 — to refuse it aloud, with a counter-offer.